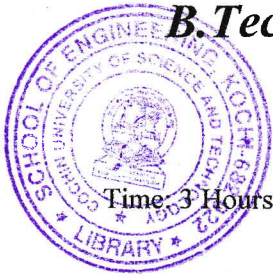


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B.Tech. Degree IV Semester Examination April 2019

ME 1402 METROLOGY AND INSTRUMENTATION (2012 Scheme)

Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Describe the methods for estimating accuracy and precision.
- (b) Briefly explain gauge maker's tolerance.
- (c) Explain the working principle of an auto collimator.
- (d) Write a note on optimeters.
- (e) Distinguish between systematic error and random error.
- (f) Briefly explain classification of measuring instruments.
- (g) Give a brief account on total radiation pyrometer.
- (h) Briefly explain the working of scintillation counter.

PART B

(4 × 15 = 60)

- II. (a) Describe the necessity and importance of metrology. Also explain system of limits and fits. (7)
- (b) Briefly explain line standard and end standard. (8)
- OR**
- III. (a) Explain hole basis and shaft basis system of representation. (7)
- (b) Write notes on: (8)
 - (i) Classification of gauges
 - (ii) Gauge materials
- IV. Explain the uses of sine bar with a neat sketch. (15)
- OR**
- V. (a) Explain the construction and working of tool maker's microscope. (7)
- (b) State the adverse effects of poor surface finish. Give any four symbols for direction of lay. (8)
- VI. (a) Describe the working of mercury in glass thermometer as a first order instrument. (7)
- (b) With diagram explain functional elements of an instrument. (8)
- OR**
- VII. (a) Explain the working of seismic instrument as a second order instrument. (7)
- (b) Discuss the generalized mathematical model of measuring systems of second order instruments. (8)
- VIII. With neat sketch explain the working of: (15)
 - (i) ORSAT's apparatus
 - (ii) Hydraulic load cell
- OR**
- IX. Briefly explain: (15)
 - (i) Bonded strain gauges
 - (ii) Strain rosettes
 - (iii) Acoustical measurements